Time-series of new Ebola BVD cases in the DRC

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Executive summary

- This report presents the daily number of new suspected and confirmed Ebola BVD cases in the DRC by date of notification, complementing the Epicentre dashboard, which currently shows latest cumulative totals only.
- Most reported cases to date have been in Ituri, particularly in Mongbalu, Bunia, and Rwampara, while other provinces and health zones have reported fewer and more sporadic cases.
- The time series were reconstructed from three sources: harmonised data compiled by Epicentre up to 25 May, official INSP data from 26 May onward, and selected unofficial updates used to improve timeliness.
- Recent gaps between high numbers of suspected cases and lower numbers of confirmed cases likely reflect delays or gaps in laboratory testing and reporting, rather than a true decline in transmission.
- Because early outbreak data have been incomplete and sometimes inconsistent across sources, these estimates should be interpreted cautiously, especially for the first days of the outbreak.

Method

Since the declaration of the 17th Ebola outbreak in DRC on 15 May, sitreps from multiple sources (INSP, WHO, MSF) and in different formats have been released. These reports have sometimes contained incomplete or conflicting figures, both across sources and over time. This reflects the challenges of establishing a reliable data pipeline at the start of an outbreak. The situation has improved with the daily publication of INSP sitreps and the release of aggregated raw data in CSV format through the INSP GitHub repository.

To reconstruct the daily time series, we combined three data sources:

1. **Harmonised data (up to 25 May)** - Data manually digitised from several sources (INSP, WHO, and MSF) and reconciled by the Epicentre Data Science team. Where available, we used reported new cases; otherwise, we calculated daily counts

from changes in cumulative totals. We have sometimes had to make arbitrary choices when several options were available; that is why these estimates should be treated with caution, especially for suspected cases (see note below).

2. **INSP data (from 26 May)** - Official INSP sitrep data accessed through the INSP GitHub repository. In some instances, the repository provides more complete information than the reports themselves. For example, sitrep n°12 reported 16 confirmed cases without specifying their origin, whereas this detail was available in the GitHub data.
3. **Unofficial sources** - Data shared directly with MSF from unofficial channels. These include newly cases reported one day before they appear in the INSP sitrep, as well as two confirmed cases in South Kivu on 26 May and that are not mentioned in the latest INSP sitrep (N°13).

For transparency, the time series identifies the source of each data point. As the data pipeline continues to improve, we expect to rely increasingly on the INSP GitHub repository as the main source, supplemented by unofficial real-time updates when available.

Definitions (from DPS Ituri)

Alert:

Any person presenting at least two of the following signs: fever, body aches, headache, physical weakness, abdominal pain, or vomiting.

Suspected case:

Any person presenting at least three of the following signs: fever, body aches, headache, physical weakness, abdominal pain, or vomiting, together with an epidemiological link.

Any case presenting with fever and a haemorrhagic syndrome (for example, epistaxis, haematemesis, or genital bleeding).

Probable case: Any person who died with fever and haemorrhage.
[Note that most INSP sitreps do not report probable cases. Deaths are usually reported as suspected or confirmed. As such we ignore probable cases for now.]

Note on the status of cases

To understand the charts below, it helps to know how cases are classified. When someone is reported (alert) and meets the definition of a suspected case, they are added to the new suspected cases for that day (D1). A sample is then taken and tested in the laboratory. When the result is available (on day D2 > D1), the person is

reclassified as either a confirmed case or a non-case, and is removed from the suspected list. This is why the total number of confirmed cases should keep increasing, while the total number of suspected cases can sometimes go down if more people are reclassified as non-case than newly reported as suspect on a given day.

New suspected cases and new confirmed cases should be interpreted differently. The number of new suspected cases reported on day D1 does not change after test results are available on day D2, because aggregated data cannot be updated retrospectively. As a result, this number mainly indicates how many test results are expected in the following days, rather than how many people are still awaiting testing at the present time. By contrast, new confirmed cases reported on a given day are people whose diagnosis has been confirmed at that time, and this status will not change. In short, “suspected” is temporary (assuming lab tests are available), while “confirmed” is final. As a result, the cumulative number of daily new suspected cases is higher than the current total of suspected cases, whereas the cumulative number of daily new confirmed cases matches the current total of confirmed cases.

Daily new cases by Province

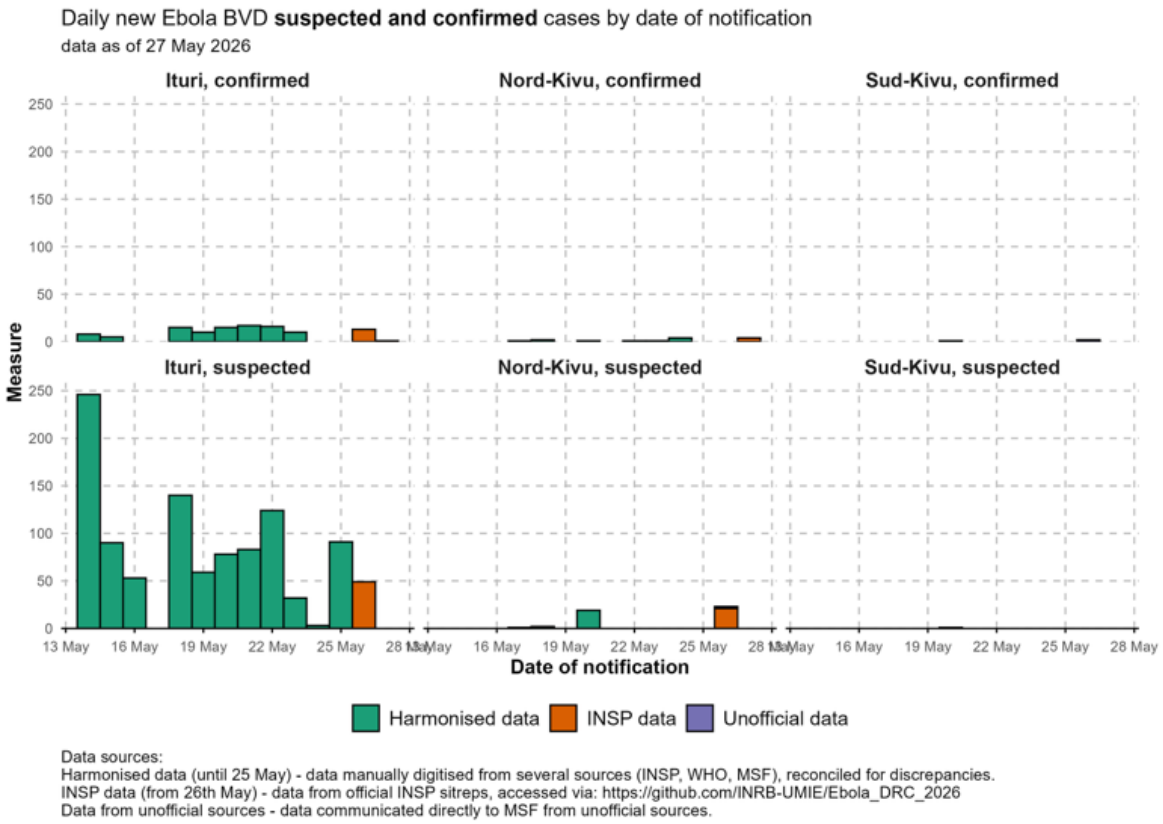


Figure 1 Daily new confirmed (upper panels) and suspected (lower panels) cases in Ituri, North Kivu, and South Kivu, by data source. All six plots use the same y-axis, showing that most daily cases were reported in Ituri. The high number of suspected cases in Ituri on 14 May reflects the first report of suspected cases investigated retrospectively.

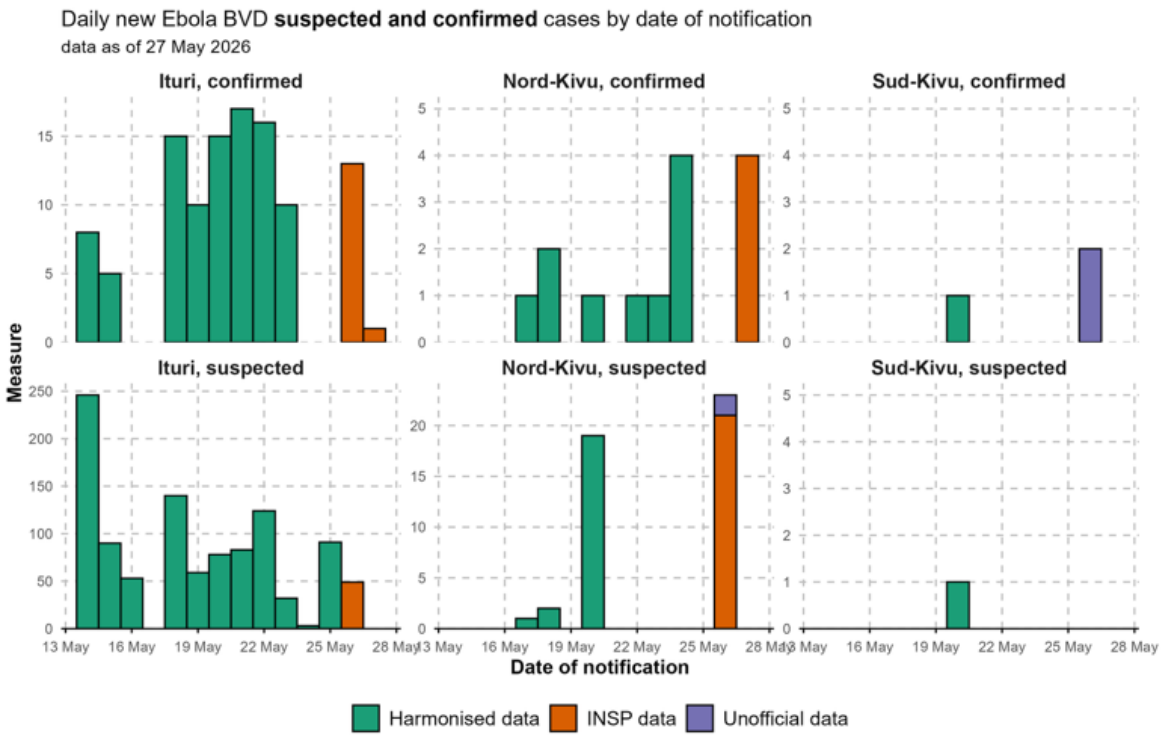


Figure 2 Same as Figure 1, but with a different y-axis for each province. This highlights the two spikes in suspected cases in North Kivu on 20 and 26 May, as well as the two confirmed cases unofficially reported in South Kivu on 26 May. Overall, the recent lack of confirmed cases despite the large number of suspected cases likely reflects delays or gaps in laboratory testing and reporting, rather than a real decline in cases.

Daily new cases by Health-Zone

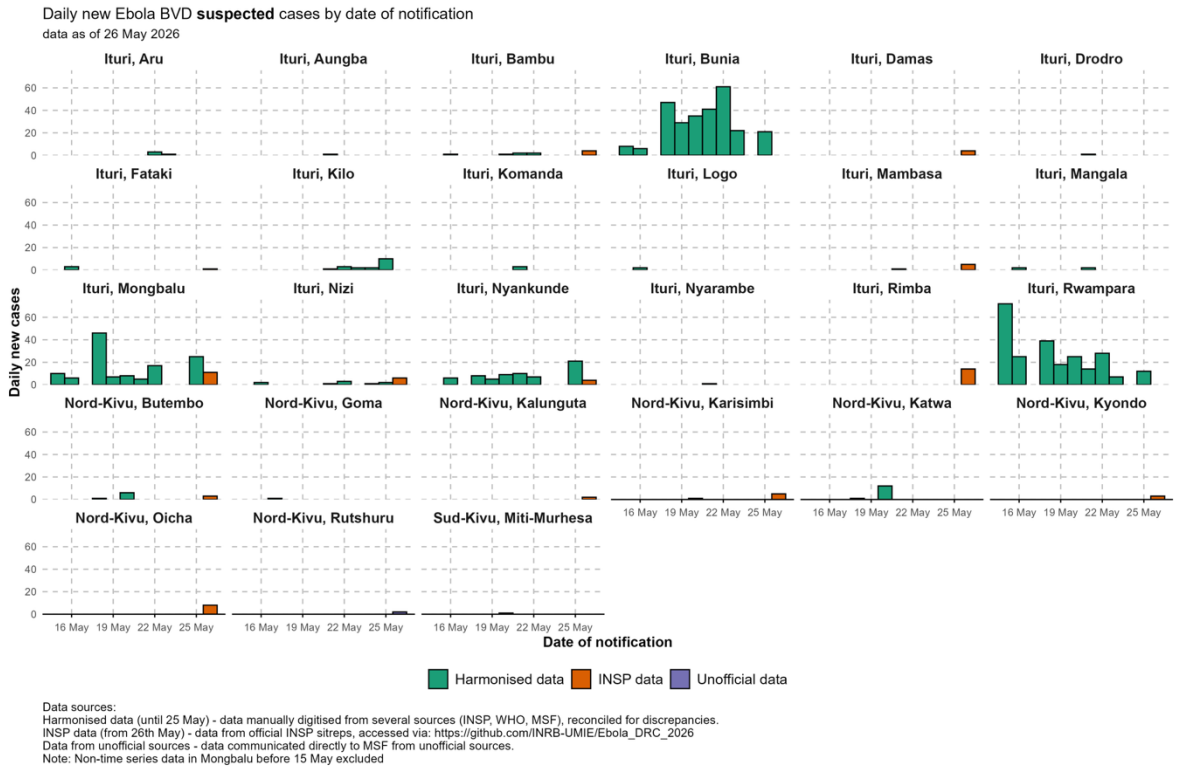


Figure 3 Daily new suspected cases in affected health-zones, by data source. All plots use the same y-axis, showing that most suspected cases were reported in Mongbalu, Bunia and Rwampara. The 246 suspected cases in Mongbalu on 14 May are excluded for the sake of visibility.

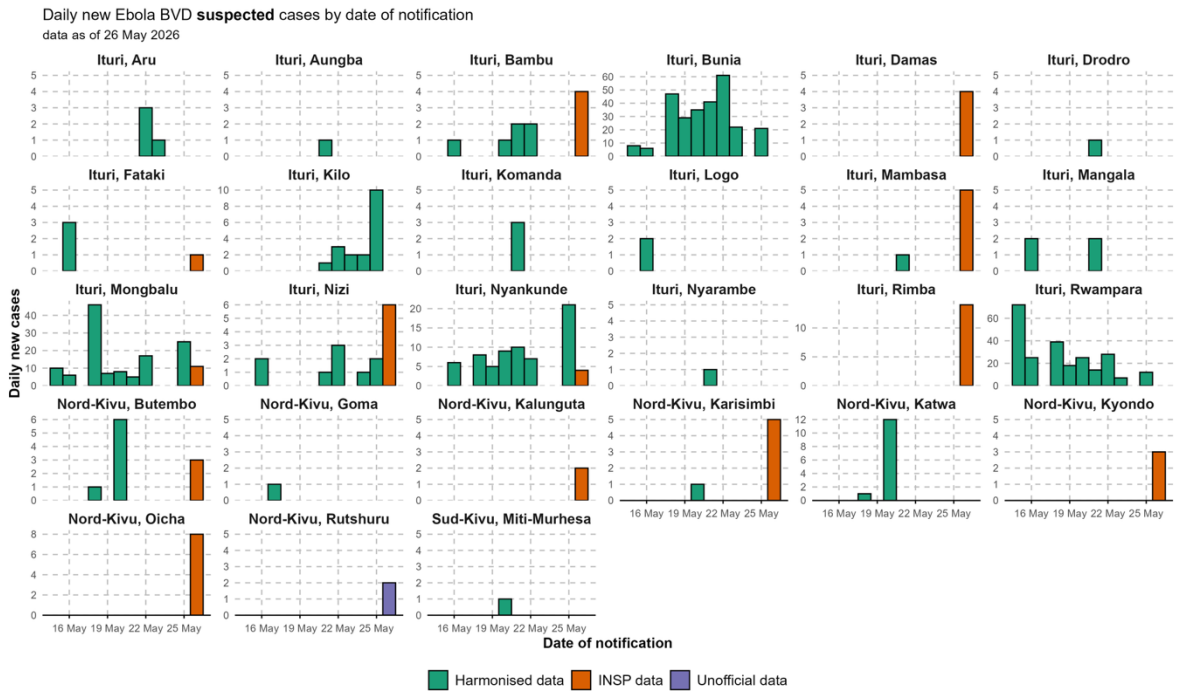


Figure 4 Same as Figure 3, but with a different y-axis for each health zone. This shows that some health-zones have reported only sporadic suspected cases, whilst others have reported a steady number of cases.

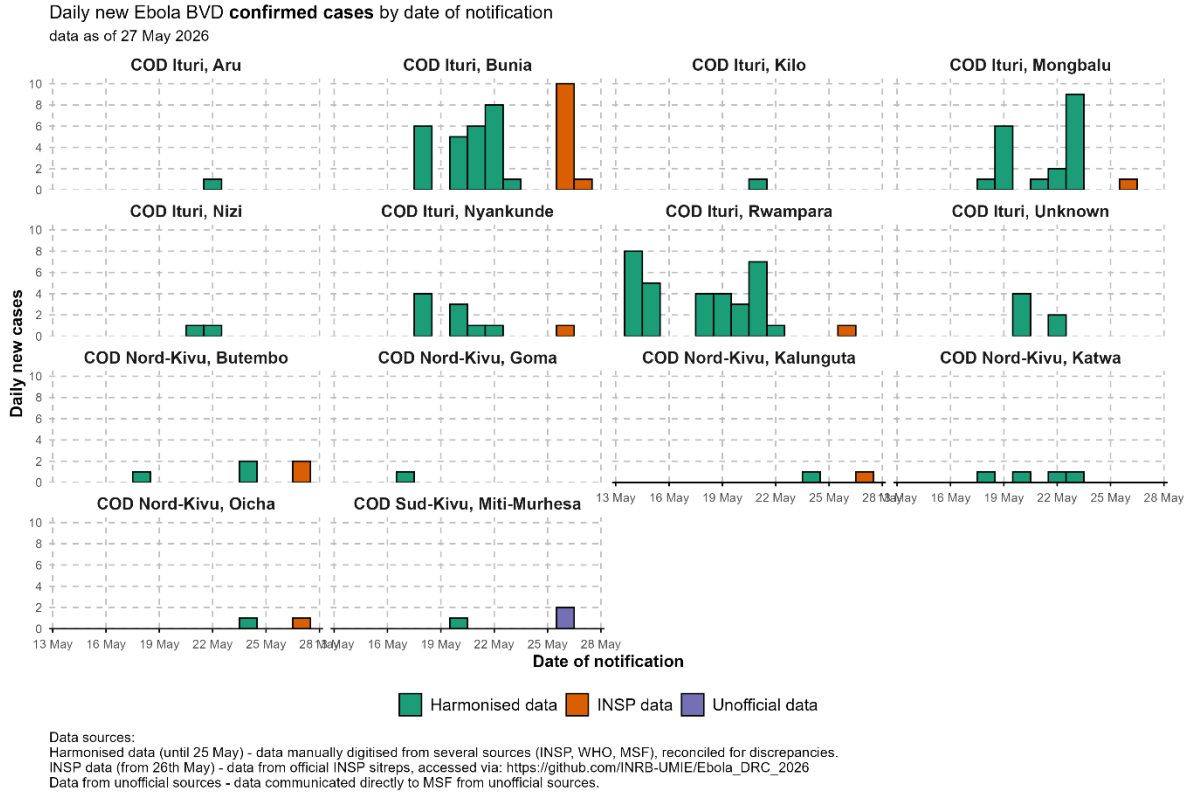


Figure 5 Same as Figure 4, but focusing on confirmed cases only. Outside Bunia, few confirmed cases have been reported in recent days across other health zones, likely reflecting delays or gaps in laboratory testing rather than a true decline in cases. Note also the two confirmed cases unofficially reported in Miti-Murhesa (South Kivu) on 26 May.